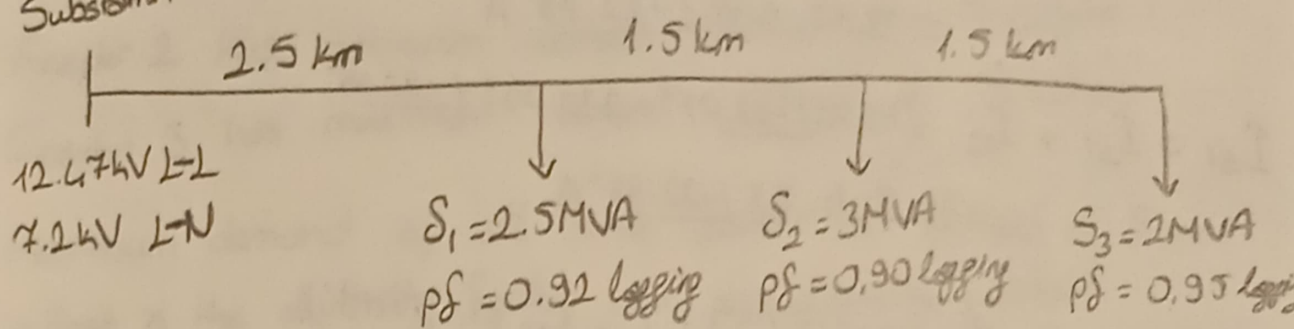


ELK406E - Distribution of Electrical Energy

Substation



Assume this three-phase feeder main with series impedance

$$Z = 0.1603 + j0.4127 \Omega/\text{km} = 0.4427 \angle 68.78^\circ \Omega/\text{km}$$

Find the percent voltage drop across the entire primary feeder for the stipulated load conditions, using the approximate method

$$Z_{S1} = 0.4427 \angle 68.78^\circ \times 2.5 = 1.10675 \angle 68.78^\circ \Omega = 0.4006 + j1.0317 \Omega$$

$$Z_{S2} = 0.4427 \angle 68.78^\circ \times 1.5 = 0.66405 \angle 68.78^\circ \Omega = 0.362 + j0.619 \Omega$$

$$Z_{S3} = 0.4427 \angle 68.78^\circ \times 1.5 = 0.66405 \angle 68.78^\circ \Omega = 0.362 + j0.619 \Omega$$

$$I = \frac{S}{\sqrt{3}V} \quad I_{L3} = \frac{2.0 \times 10^6}{\sqrt{3} \times 12.47 \times 10^3} \angle -\cos^{-1}(0.95)^\circ = 92.60 \angle -18.19^\circ \text{ A}$$

$$I_{L2} = \frac{3.0 \times 10^6}{\sqrt{3} \times 12.47 \times 10^3} \angle -\cos^{-1}(0.90)^\circ = 138.89 \angle -25.84^\circ \text{ A}$$

$$I_{L1} = \frac{2.5 \times 10^6}{\sqrt{3} \times 12.47 \times 10^3} \angle -\cos^{-1}(0.92)^\circ = 115.76 \angle -23.07^\circ \text{ A}$$

These are the currents drawn by each load

$$I_{23} = I_{13} = 92.60 \angle -18.19^\circ \text{ A}$$

$$\begin{aligned} I_{12} &= I_{12} + I_{23} = 138.89 \angle -25.84^\circ + 92.60 \angle -18.19^\circ \\ &= 230.995 \angle -22.78^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} I_{s1} &= I_{11} + I_{12} = 115.74 \angle -23.07^\circ + 230.995 \angle -22.78^\circ \\ &= 346.73 \angle -22.88^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} VD_{23} &= I_{23} R_{23} \cos(18.19^\circ) + I_{23} X_{23} \sin(18.19^\circ) \\ &= 92.60 \times 0.362 \cos(18.19^\circ) + 92.60 \times 0.619 \sin(18.19^\circ) \\ &= 31.85 + 17.89 = 49.74 \text{ V} \end{aligned}$$

$$\begin{aligned} VD_{12} &= I_{12} R_{12} \cos(22.78^\circ) + I_{12} X_{12} \sin(22.78^\circ) \\ &= 230.995 \times 0.362 \cos(22.78^\circ) + 230.995 \times 0.619 \sin(22.78^\circ) \\ &= 77.038 + 55.36 = 132.46 \text{ V} \end{aligned}$$

$$\begin{aligned} VD_{s1} &= I_{s1} R_{s1} \cos(22.88^\circ) + I_{s1} X_{s1} \sin(22.88^\circ) \\ &= 346.73 \times 0.4006 \cos(22.88^\circ) + 346.73 \times 1.0317 \sin(22.88^\circ) \\ &= 127.97 + 139.08 = 267.05 \text{ V} \end{aligned}$$

$$\Sigma VD = 49.74 + 132.46 + 267.05 = 449.25 \text{ V}$$

$$\frac{449.25}{7200} = 6.24 \%$$

Longitudinal Voltage Drop, $\Delta U = R I_p + X I_q$ affects magnitude.

Transversal Voltage Drop, $\delta U = X I_p - R I_q$ affects phase angle.

Example

A substation has three outgoing feeders

Feeder 1 has maximum demand 10 MW at 10.00 pm

Feeder 2 has maximum demand 12 MW at 7.00 pm

Feeder 3 has maximum demand 15 MW at 9.00 pm

Maximum demand of all three feeder is 33 MW at 8 pm.

What is the diversity factor and simultaneity factor?

$$\text{Diversity factor} = \frac{\text{Sum of individual max demand}}{\text{max demand on station}} = \frac{37}{33}$$

$$\text{Simultaneity factor} = \frac{1}{\text{diversity factor}} = \frac{1}{1.12} = 89.3\%$$

Example

There are three consumers of electricity having different load requirements at different times.

Consumer 1 has a maximum demand of 5 kW at 6 pm, 3 kW demand at 7 pm and daily load factor of 20%.

Consumer 2 has a maximum demand of 5 kW at 11 am, 2 kW demand at 7 pm, and average load of 1200 W.

Consumer 3 has an average load of 1 kW and has a maximum demand of 3 kW at 7 pm.

Determine;

a. The diversity factor

b. The load factor and average load of each consumer

c. The average load and load factor of the combined load

Maximum demand of the system is at 7 pm with 8 kW

$$\text{Diversity factor} = \frac{5+5+3}{8} = 1.625$$

$$\text{Average load} = \text{load factor} \times \text{peak load}$$

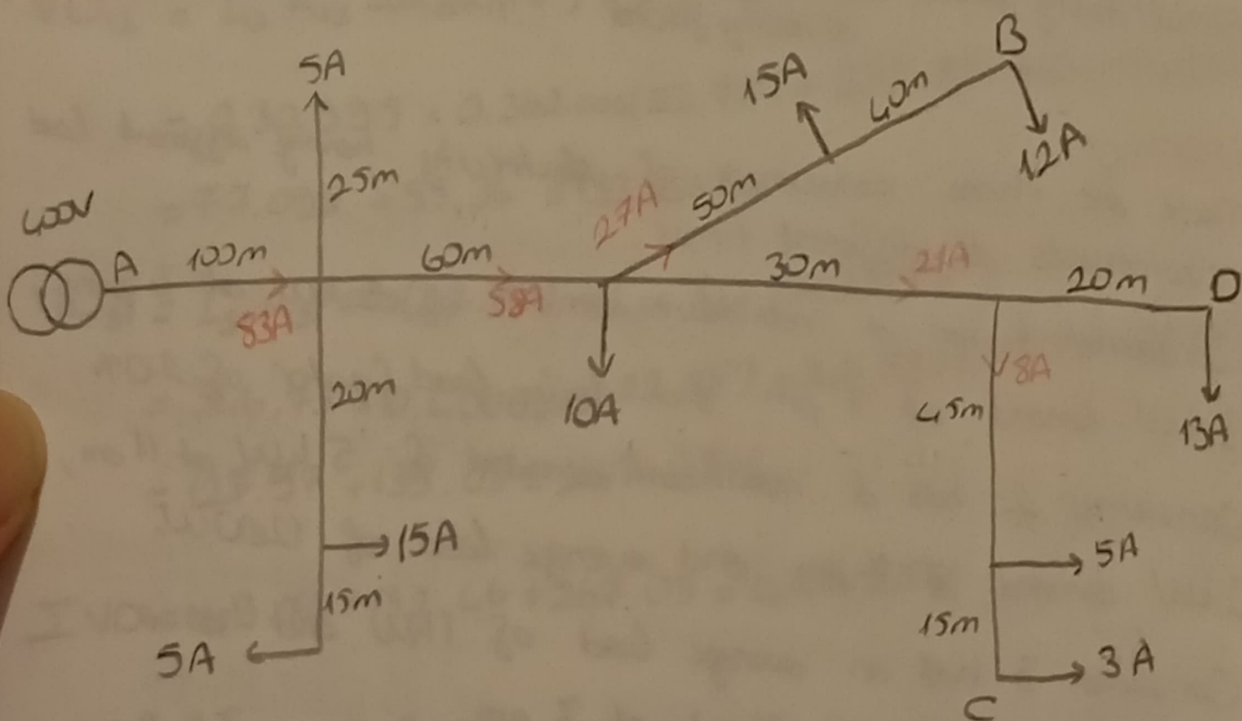
$$P_{avg1} = 0.2 \times 5 = 1 \text{ kW} \quad P_{avg2} = 0.8 \times 5 = 1200 \text{ W} \rightarrow 24\%$$

$$P_{avg3} = 0.8 \times 3 = 1000 \text{ W} \rightarrow 33.3\%$$

$$\text{Combined average load} = 1 + 1.2 + 1 = 3.2 \text{ kW}$$

$$\text{Combined load factor} = \frac{\text{combined average load}}{\text{max demand on station}} = \frac{3.2}{8} \times 100 = 40\%$$

Example



Consider the three-phase three-wire 400 V secondary system with balanced loads. Determine the following:

The critical node where the maximum voltage drop occur.

Calculate the minimum cross-sectional area of the conductor.

Calculate the total line real power loss.

$$\chi_{cu} = 56 \text{ m}/\Omega \text{ mm}^2$$

$$\Delta U_{AB} = R \cdot I_p = \frac{83 \cdot 100 + 58 \cdot 60 + 27 \cdot 50 + 12 \cdot 40}{56 \cdot q} = \frac{13610}{56 \cdot q}$$

$$= \frac{243,03}{q}$$

$$\Delta U_{AC} = R \cdot I_p = \frac{83 \cdot 100 + 58 \cdot 60 + 30 \cdot 21 + 8 \cdot 45 + 3 \cdot 15}{56 \cdot q}$$

$$= \frac{12815}{56 \cdot q}$$

$$\Delta U_{AD} = R \cdot I_p = \frac{83 \cdot 100 + 58 \cdot 60 + 21 \cdot 30 + 13 \cdot 20}{56 \cdot q} = \frac{12670}{56 \cdot q}$$

$$V_{ph} = \frac{V_{LL}}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230V$$

Voltage limit is 5% so maximum voltage drop is $\Delta U_{max} = 230 \cdot \frac{5}{100} = 11,54V$

$$\Delta U_{AB} = \frac{243,03}{q} = 11,54 \rightarrow q = 21,06 \text{ mm}^2$$

$$P_{loss} = 3 R I^2$$

we choose 25 mm² from the manufacturer

$$P_{loss} = \frac{3}{56 \cdot 25} (100 \cdot 83^2 + 25 \cdot 5^2 + 20 \cdot 20^2 + 15 \cdot 5^2 + 60 \cdot 58^2 + 50 \cdot 27^2 + 40 \cdot 12^2 + 30 \cdot 21^2 + 45 \cdot 8^2 + 15 \cdot 3^2 + 20 \cdot 13^2)$$

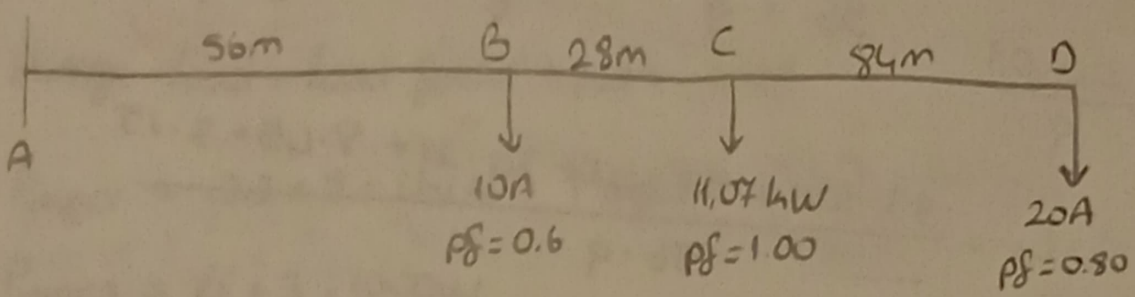
$$P_{loss} = \frac{3 \cdot 961575}{56 \cdot 25} = 2060W = 2,06kW$$

We use phase voltage for voltage drop limits because at the distribution level, most loads operate on single-phase power.



Example

231/400V



$q = 10 \text{ mm}^2$

$\chi_{cu} = 56 \text{ m} / \Omega \text{ mm}^2$

- a) Calculate the voltages at A, B, D
- b) Calculate total line real power loss.

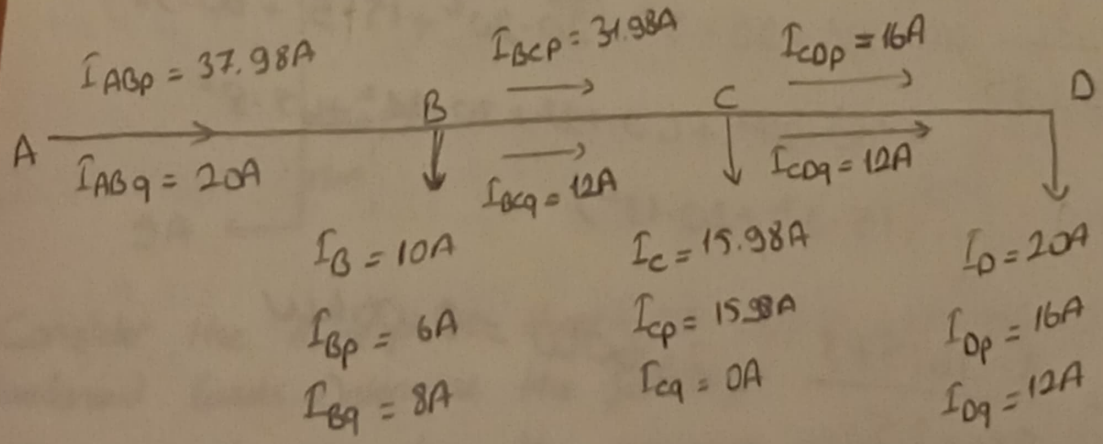
a)

$I_0 = 10$ $I_{Bp} = 10 \cdot \cos \theta = 10 \cdot 0.6 = 6 \text{ A}$

$I_{Bq} = 10 \cdot \sin \theta = 10 \cdot 0.8 = 8 \text{ A}$

$P_C = 11.07 \text{ kW} = \sqrt{3} \cdot V_L \cdot I_{cp}$ $I_{cp} = 15.98 \text{ A}$ $I_{cq} = 0 \text{ A}$

$I_D = 20$ $I_{Dp} = 16 \text{ A}$ $I_{Dq} = 12 \text{ A}$



$V_D = R \cdot I_p$ $\Delta V_{AB} = \frac{56}{56 \cdot 10} \cdot 37.98 = 3.8 \text{ V}$

$R = \frac{2}{\chi \cdot A}$
cross sectional area

$\Delta V_{BC} = \frac{28}{56 \cdot 10} \cdot 31.98 = 1.6 \text{ V}$

$\Delta V_{CD} = \frac{84}{56 \cdot 10} \cdot 16 = 2.4 \text{ V}$

$$V_D = V_C - \Delta V_{CD} = 231 - 2.4 = 228.6 \text{ V}$$

$$V_B = V_C + \Delta V_{BC} = 231 + 1.6 = 232.6 \text{ V}$$

$$V_A = V_B + \Delta V_{AB} = 232.6 + 3.8 = 236.4 \text{ V}$$

$$b) I_{AB} = \sqrt{37.98^2 + 20^2} = 42.924 \text{ A}$$

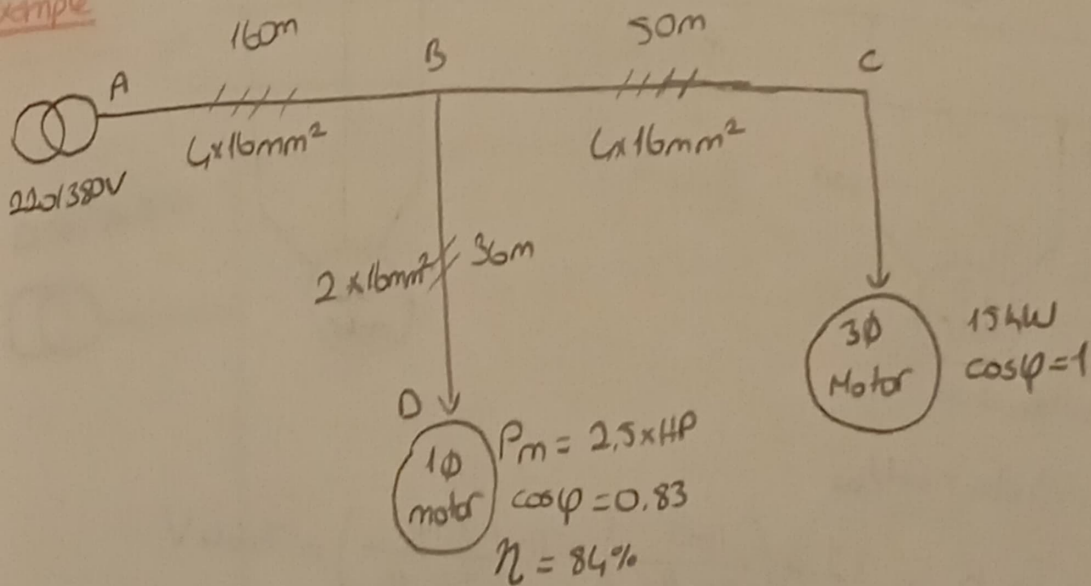
$$I_{BC} = \sqrt{31.98^2 + 12^2} = 34.157 \text{ A}$$

$$I_{CD} = \sqrt{16^2 + 12^2} = 20 \text{ A}$$

$$P_{\text{Loss}} = 3RI^2 = \frac{3}{56 \cdot 10} [56 \cdot 42.924^2 + 28 \cdot 34.157^2 + 84 \cdot 20^2]$$

$$P_{\text{Loss}} = 907.74 \text{ W}$$

Example



a) Calculate voltage at point D. Check if voltage drop is in tolerable limit (5%) HP = 0.736kW $\chi_{cu} = 56 \text{ m} / 2 \text{ mm}^2$

b) Calculate power loss on each pair

$$P_m = 2.5 \times 0.736 \text{ kW} = 1.84 \text{ kW} \quad \eta = \frac{P_m}{P_e}$$

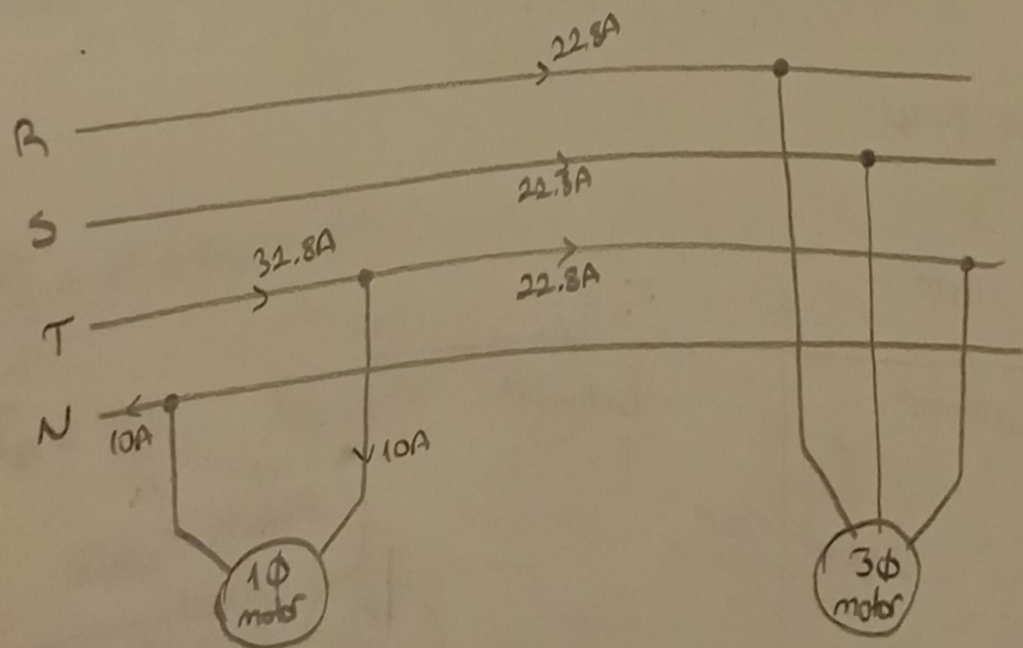
$$0.84 = \frac{1.84}{P_e} \quad P_e = 2.2 \text{ kW}$$

$$1\phi \text{ motor power formula } P = V_{L-n} \cdot I \cdot \cos\phi = V \cdot I_p$$

$$I_{pd} = P_e / V_{L-n} = 2200 \text{ W} / 220 \text{ V} = 10 \text{ A}$$

$$\text{For } 3\phi \text{ motor } P_c = \sqrt{3} V_{L-L} \cdot I_{pc}$$

$$I_{pc} = \frac{P}{\sqrt{3} \cdot V_{L-L}} = \frac{15 \cdot 10^3}{\sqrt{3} \cdot 380} = 22.8 \text{ A}$$



For voltage drop

$$\Delta U_{ad} = \Delta U_{ab} + \Delta U_{bd}$$

$$\Delta U_{ab} = I_p \cdot R = 32.8 \left(\frac{160}{56.16} \right) + 10 \left(\frac{160}{56.16} \right) = 7.64 \text{ V}$$

$$\Delta U_{bd} = 2 \cdot 10 \cdot \left(\frac{36}{56.16} \right) = 2.14 \text{ V}$$

$$\Delta U_{bd} = 9.78 \text{ V} \quad \Delta U_{max} = \frac{5}{100} \cdot 220 = 11 \text{ V} \quad \checkmark$$

$$P_{loss} = 3RI^2$$

$$\text{for } bc = 3 \left(\frac{50}{56.16} \right) \cdot 22.8^2 = 87.63 \text{ W}$$

$$I_{qd} = I \cdot \sin \varphi = I_{pd} \cdot \tan \varphi = 6.72 \text{ A}$$

→ because we have
found I value using P_2 formula

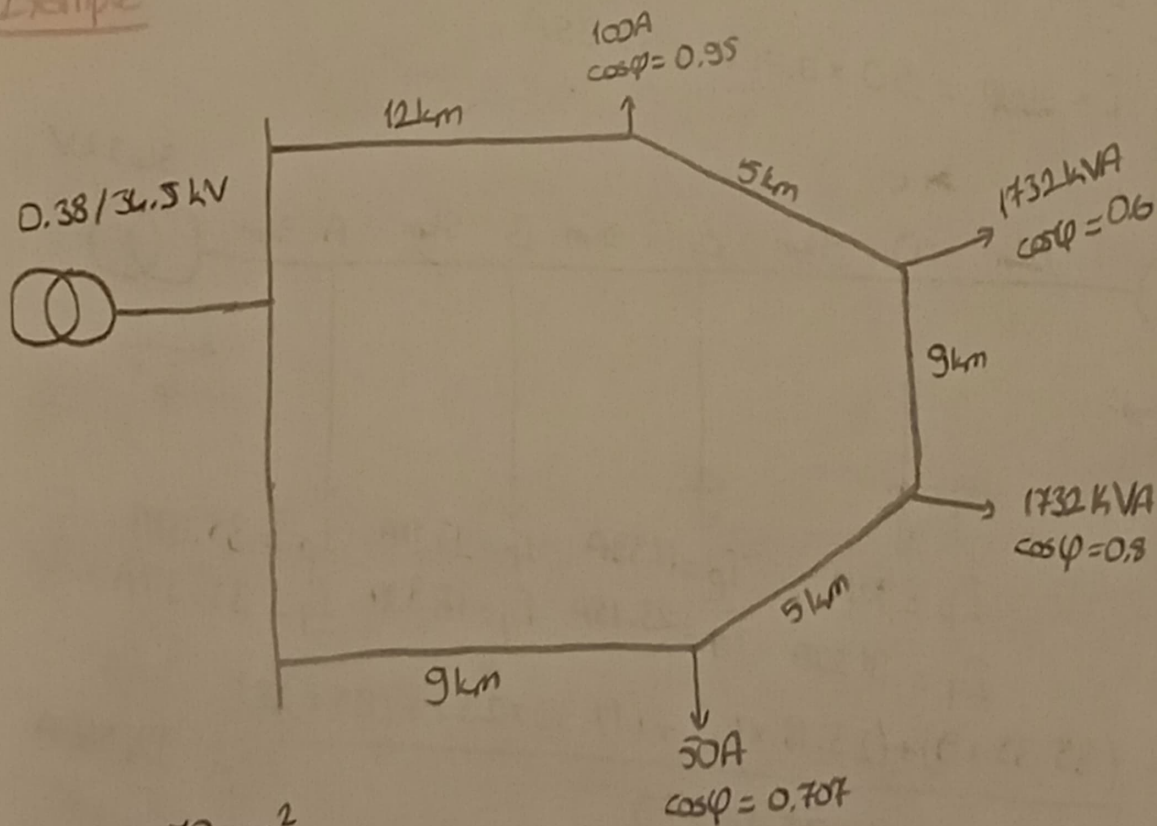
$$I_d = \sqrt{6.72^2 + 10^2} = 12.05 \text{ A}$$

$$\text{for } ab = 2 \left(\frac{160}{56.16} \right) \cdot 22.8^2 + (32.8^2 + 6.72^2) \left(\frac{160}{56.16} \right) + 12.05^2 \left(\frac{160}{56.16} \right) = 411.76 \text{ W}$$

$$\text{for } bd = 2 \left(\frac{36}{56.16} \right) 12.05^2 = 49.48 \text{ W}$$

$$P_{loss \text{ total}} = 411.76 + 87.63 + 49.48 = 548.87 \text{ W}$$

Example



$$q = 3 \times 70 \text{ mm}^2$$

$$\chi_{cu} = 56 \text{ m}/\Omega \text{ mm}^2$$

$$\text{unit reactance} = 0.2 \Omega / \text{km}$$

a) Find the critical node where the maximum longitudinal voltage drop occurs.

b) Calculate the transversal voltage drop at the critical node

$$I_{pd} = I \times \cos \varphi = 100 \times 0,95 = 95 \quad \varphi = 53^\circ$$

$$I_{qd} = 100 \times 0,3112 = 31,22 \text{ A}$$

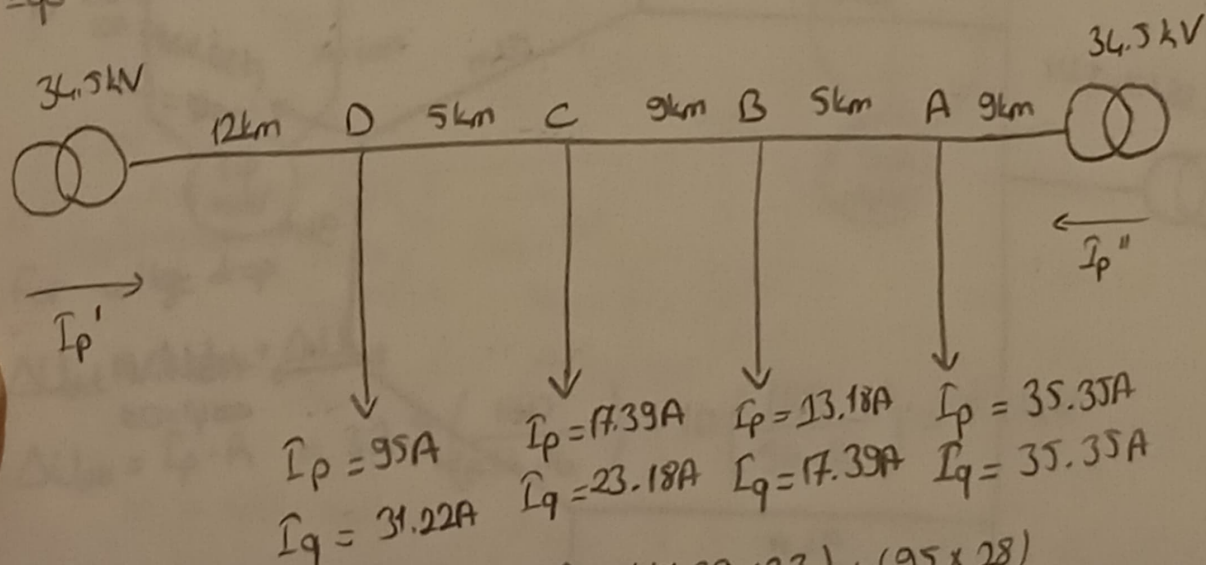
$$I_c = \frac{S}{\sqrt{3} \cdot V} \rightarrow I_c = 28,98 \text{ A}$$

$$I_{pc} = 28,98 \cdot 0,6 = 17,39 \text{ A} \quad I_{qc} = 28,98 \times 0,8 = 23,18 \text{ A}$$

$$I_{pb} = \frac{S \cos \varphi}{\sqrt{3} V} \rightarrow I_{pb} = 23,18 \text{ A} \quad I_{qb} = 17,39 \text{ A}$$

$$I_{ps} = I \times \cos \varphi = 50 \times 0,707 = 35,35 \text{ A} \quad \varphi = 45^\circ$$

$$I_{qs} = I \times \sin \varphi = 50 \times 0,707 = 35,35 \text{ A}$$



$$I_{p'} = \frac{(35,35 \times 9) + (23,18 \times 14) + (17,39 \times 23) + (95 \times 28)}{12 + 5 + 9 + 5 + 9} = 92,56 \text{ A}$$

$$I_{p\text{total}} = I_{p'} + I_{p''} = 95 + 17,39 + 23,18 + 35,35 = 170,92$$

$$I_{p''} = 78,354 \text{ A}$$

$$I_q'' = \frac{(31.22 \times 12) + (23.18 \times 12) + (17.39 \times 26) + (35.35 \times 31)}{40} = 57.92A$$

$$I_{q\text{total}} = 31.22 + 23.18 + 17.39 + 35.35 = 107.14A$$

$$I_q' = 49.22A \quad \text{Node D}$$

$$\Delta U_d = R I_p + X I_q = \frac{12000}{56.70} \cdot 92.56 + 0.2 \cdot 12 \cdot 49.22$$

$$\Delta U_d = 401.46V$$

$$\% \varepsilon = \frac{401.46}{\frac{34.5 \times 10^3}{\sqrt{3}}} \times 100 = 2\%$$

Node C

$$\Delta U_c = R \cdot I \cdot \cos \varphi + X \cdot I \cdot \sin \varphi = \frac{1000}{56.70} (12 \cdot 92.56 - 5 \cdot 2.44) + 0.2 (12 \cdot 49.22 + 5 \cdot 18)$$

$$\Delta U_c = 416.358$$

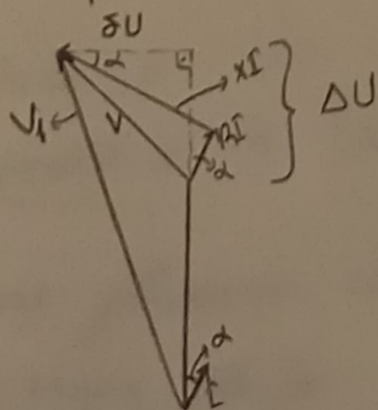
$$\% \varepsilon = \frac{416.358}{\frac{34.5 \times 10^3}{\sqrt{3}}} \times 100 = 2.09\%$$

$$\delta U_c = X I \cos \varphi - R I \sin \varphi = 0.2 (12 \cdot 92.56 - 5 \cdot 2.44) - \frac{1000}{56.70} (12 \cdot 49.22 + 5 \cdot 18)$$

$$\delta U_c = 46.074V$$

$$E = \sqrt{(V + \Delta U_c)^2 + \delta U^2} = \frac{(34.5k)^2}{(\sqrt{3})^2} = \left(\frac{V_c}{\sqrt{3}} + 416.358\right)^2 + 46.074^2$$

$$V_c = 33.78kV$$



$$\Delta U = R I_p + X I_q \quad (\text{Longitudinal})$$

$$\delta U = X I_p - R I_q \quad (\text{Transversal})$$

Faults and Short Circuits

Phase-to-phase or phase-to-ground contact

May be "bolted" (zero impedance) or have some impedance

Maximum I_{sc} used for equipment rating

Minimum I_{sc} used for fuse/protection settings

Electric arc at point of fault causes destruction

Thermal effect: overheating, deterioration of the insulation

Electrodynemic effect: When a fault current flows through a conductor, it generates a strong magnetic field. This field interacts with the current itself, producing a mechanical force. The result is repulsion or attraction between parallel conductors or between parts of busbars.

This force is proportional to the square of the current.

(Laplace's Force)

$$F = \vec{I} \cdot (\vec{L} \times \vec{B}) \quad \frac{\mu_0 I}{2\pi r}$$

$$F \propto I^2$$

Symmetrical Component

Any unbalanced 3 ϕ system can be represented as the sum of three balanced systems:

1. Positive Sequence (balanced, same order)
2. Negative Sequence (balanced, reverse order)
3. Zero Sequence (identical in all phases)

$$I_1 = \frac{1}{3} (I_A + \alpha I_B + \alpha^2 I_C) \quad (\text{positive-sequence})$$

$$I_2 = \frac{1}{3} (I_A + \alpha^2 I_B + \alpha I_C) \quad (\text{negative-sequence})$$

$$I_0 = \frac{1}{3} (I_A + I_B + I_C) \quad (\text{zero-sequence})$$

E : System power supply phase-to-neutral voltage

Z : Fault impedance

$Z_{(1)}$: System positive-sequence impedance

$Z_{(2)}$: System negative-sequence impedance

$Z_{(0)}$: System zero sequence impedance

Given
values

I_1, I_2, I_3 : Real phase currents at the fault point

V_1, V_2, V_3 : Real phase-to-neutral voltages at the fault point

$I_{(1)}, I_{(2)}, I_{(0)}$: Positive-sequence, Negative-sequence, zero-sequence current system at the fault point

$V_{(1)}, V_{(2)}, V_{(0)}$: Positive-sequence, negative-sequence, zero-sequence voltage system at the fault point

$$\vec{V}_1 = \vec{V}_{(1)} + \vec{V}_{(2)} + \vec{V}_{(0)}$$

$$\vec{V}_{(1)} = \vec{E} - Z_{(1)} \vec{I}_{(1)}$$

$$\vec{V}_2 = \alpha^2 \vec{V}_{(1)} + \alpha \vec{V}_{(2)} + \vec{V}_{(0)}$$

$$V_{(2)} = -Z_{(2)} \vec{I}_{(2)}$$

$$\vec{V}_3 = \alpha \vec{V}_{(1)} + \alpha^2 \vec{V}_{(2)} + \vec{V}_{(0)}$$

$$\vec{V}_{(0)} = -Z_{(0)} \vec{I}_{(0)}$$

$$\vec{I}_1 = \vec{I}_{(1)} + \vec{I}_{(2)} + \vec{I}_{(0)} \quad \vec{I}_2 = \alpha^2 \vec{I}_{(1)} + \alpha \vec{I}_{(2)} + \vec{I}_{(0)}$$

$$\vec{I}_3 = \alpha \vec{I}_{(1)} + \alpha^2 \vec{I}_{(2)} + \vec{I}_{(0)}$$

These are the equations governing the fault point, depending on the type of fault.

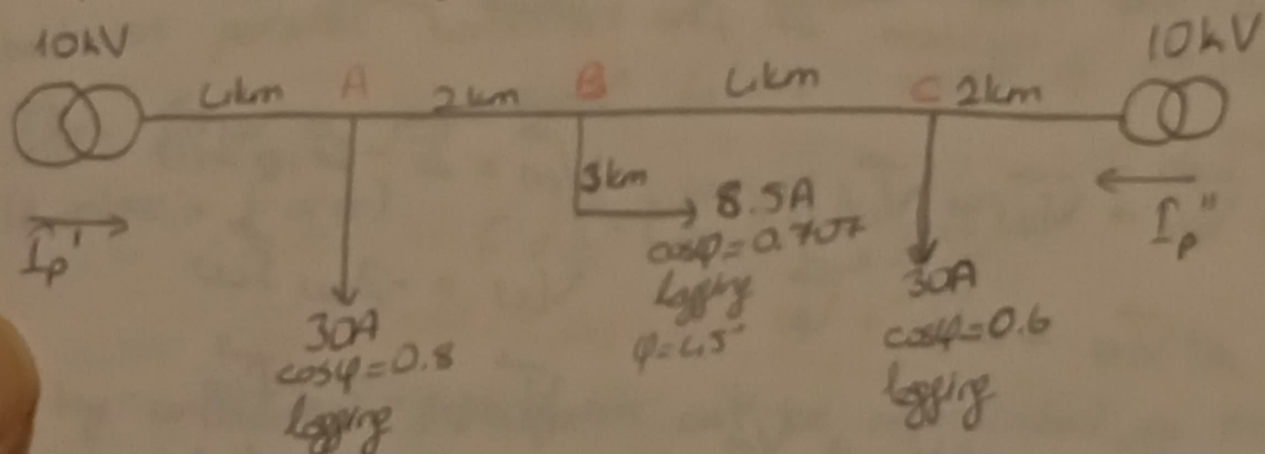
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12 oku saat

A three phase 10kV distribution system is fed at both ends with equal voltage. The unit reactance of the conductor is $0.4 \Omega/\text{km}$

- Find the current distribution in the system and determine the critical node where maximum voltage drop occurs.
- If the maximum allowable voltage drop (vertical voltage drop) is 8%, calculate the minimum cross-sectional area of the conductor and select a standard cross-sectional area.
- Check the thermal limit of the conductor (ampacity) by using the table given. $X_{cu} = 56 \text{ m}^2/\text{mm}^2$



for A

$$I_p = 30 \times 0.8 = 24 \text{ A}$$

$$I_q = 30 \times 0.6 = 18 \text{ A}$$

for C

$$I_p = 30 \times 0.6 = 18 \text{ A}$$

$$I_q = 30 \times 0.8 = 24 \text{ A}$$

for B

$$I_p = 8.5 \cos(45^\circ) = 6.01 \text{ A}$$

$$I_q = 8.5 \sin(45^\circ) = 6.01 \text{ A}$$

$$I_p' = \frac{[(18 \times 2) + (6.01 \times 9) + (24 \times 8)]}{2 + 4 + 3 + 2 + 1}$$

$$= 18.8 \text{ A}$$

$$I_{p\text{total}} = 24 + 6 + 18 = 48 \text{ A} \quad I_{p''} = I_{p\text{total}} - I_{p'} = 29,2 \text{ A}$$

$$I_{q'} = \frac{[(24 \times 2) + (6,01 \times 9) + (18 \times 8)]}{4 + 2 + 3 + 4 + 2} = 16,4 \text{ A} \quad I_{q\text{total}} = 48 \text{ A}$$

$$I_{q''} = I_{q\text{total}} - I_{q'} \Rightarrow 48 - 16,4 = 31,6 \text{ A}$$

Node A has the maximum voltage drop

$$\varepsilon (\text{percent voltage regulation}) = \frac{\Delta U}{\frac{10 \text{ kV}}{\sqrt{3}}} = 8\%$$

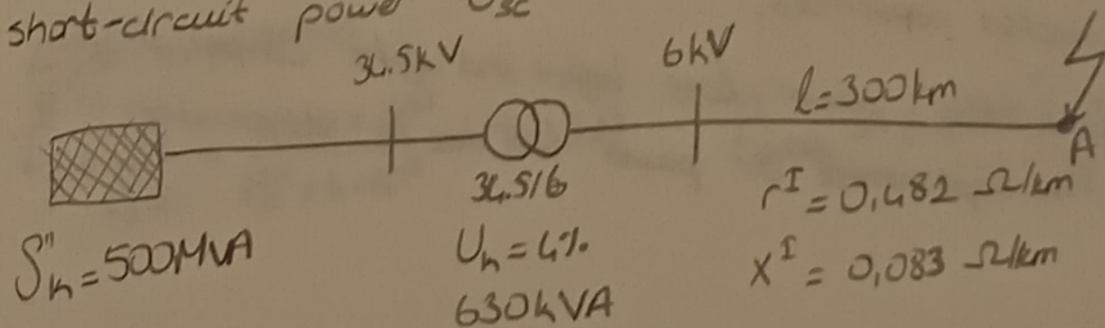
$$\Delta U = 4,61,88 \text{ V} \quad R = \frac{l}{\sigma q}$$

$$\Delta U = R I \cos \phi + X I \sin \phi \quad X = 0,4 \times 4 \text{ km} = 1,6 \Omega$$

$$= \frac{4000}{q \times 56} \times 18,8 + 16,4 \times \frac{0,4 \times 4}{X} \quad q = 3,08 \text{ mm}^2$$

Question 2

When a three-phase fault occurs at A, calculate the maximum short-circuit current and three-phase symmetrical short-circuit power S_{sc}



$$Z = 0,04 \times \frac{(6 \times 10^3)^2}{630 \times 10^3} = 2,28 \Omega \quad (\text{Transformer})$$

$$Z = \frac{U_{\text{nominal}}^2}{S} = \frac{(6 \times 10^3)^2}{500 \times 10^6} = 0,072 \Omega \quad (\text{Network})$$

$$r' = 300 \times 0,482 = 0,1446 \Omega$$

$$x' = 300 \times 0,083 = 0,0246 \Omega$$

$$Z_{total} = \left[(2,28 + 0,072 + 0,0246) j^2 + 0,1446^2 \right]^{1/2} = 2,38 \Omega$$

$$I_{sc} = \frac{U_{sc}}{\sqrt{3} \cdot Z} = \frac{6000}{\sqrt{3} \times 2,38} = 1155 A$$

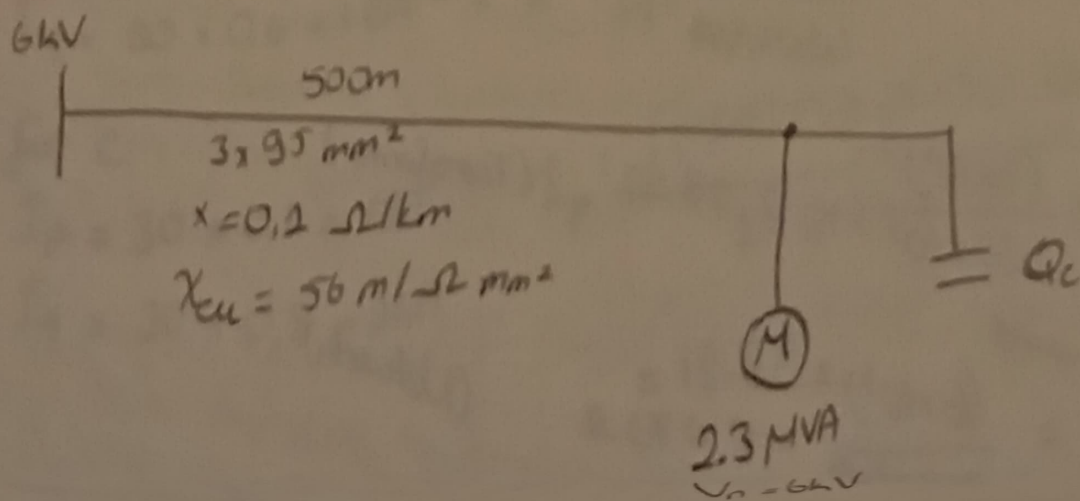
$$S_{sc} = \sqrt{3} \cdot 1155 \cdot 6 kV = 15.1 MVA$$

Question 3

In a thermal power plant, a cooling pump is supplied from a 6 kV bus. When motor operates at its rated power, its power factor is 0,8, lagging. Delta connected capacitor banks will be connected in parallel with the motor.

a) Calculate the necessary compensation power (Q_c) to correct the power factor to 0,95 and the capacitance value of the delta connected capacitors.

b) Calculate the power loss before and after compensation. Express the improvement on power loss due to compensation in %



$$S_{\text{motor}} = 2.3 \text{ MVA} \quad \cos \phi = 0.8 \quad \sin \phi = 0.6 \quad \phi = 36.87^\circ$$

$$P = 1.84 \text{ MW} \quad Q = 1.38 \text{ MVAR}$$

$$\text{After compensation} \quad \cos \phi_2 = 0.95 \quad \phi_2 = 18.19^\circ$$

$$Q_c = P(\tan \phi_1 - \tan \phi_2) = 1.84(\tan(36.87^\circ) - \tan(18.19^\circ)) \\ = 775 \text{ kVAR}$$

For Δ connected capacitors

$$Q_c = 3 V_c^2 \omega C_\Delta \quad C_\Delta = \frac{775 \times 10^3}{3(6 \times 10^3)^2 2\pi 50} = 22.8 \mu\text{F}$$

$$R = \frac{l}{\kappa q} = \frac{500}{56.95} = 0.084 \Omega \quad X_{\text{line}} = 0.5 \times 0.2 = 0.1 \Omega$$

Before compensation

$$I_1 = \frac{S_1}{\sqrt{3} V} = \frac{2.3 \times 10^6}{\sqrt{3} \times 6 \times 10^3} = 221.3 \text{ A}$$

$$P_{\text{loss}} = 3 \cdot (221.3)^2 \cdot 0.084 = 13.81 \text{ kW}$$

After compensation

$$I_2 = \frac{S_2}{\sqrt{3} V} = \frac{1936 \times 10^3}{\sqrt{3} \times 6 \times 10^3} = 186.3 \text{ A} \quad P_{\text{loss}_2} = 3 \times (186.3)^2 \cdot 0.084 = 9.79 \text{ kW}$$

$$\frac{13.81 - 9.79}{13.81} = \underline{\underline{\% 29.1}}$$